

STATIC TESTING DOCUMENT

1. Static Testing Review (Requirement-by-Requirement Analysis)

3.1 Customer Registration & Authentication

FR1 – Unique Email Registration

Issues Found:

- Email format validation not defined.
 - No description of error message when email already exists.
 - No clarification if uniqueness is checked in real-time or after form submission.
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FR2 – Password Format Validation

Issues Found:

- No password rules provided (min length, uppercase, numbers...).
 - No examples of valid or invalid passwords.
 - Not stated whether password requirements are shown to the user.
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FR3 – JWT Authentication

Issues Found:

- Token expiration time not defined.
 - No refresh token mechanism mentioned.
 - No expected system behavior after token expiration.
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FR4 – Password Reset

Issues Found:

- Reset method not defined (email link? OTP?).
 - No expiry time for reset link/OTP.
 - No identity verification steps provided.
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3.2 Product Management

FR5 – Add Product

Issues Found:

- No input validation rules (max length, formats, ranges).
 - SKU uniqueness not mentioned.
 - No error handling cases defined.
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FR6 – Update Inventory & Status

Issues Found:

- Product status values not listed (active/inactive/out-of-stock).
 - No mention of how inventory updates reflect to customers.
 - No role-based access description.
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FR7 – View Active Products

Issues Found:

- No pagination rules.
 - No sorting rules defined.
 - What happens if no active products exist?
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FR8 – Product Search & Filtering

Issues Found:

- Search type not specified (full-text, keyword).
- Combination of filters not clarified.
- Expected behavior for “no results found” missing.

3.3 Cart & Checkout

FR9 – Add/Remove Items

Issues Found:

- Quantity rules missing.
- No scenario for out-of-stock items already in the cart.
- Guest cart usage not defined.

FR10 – Auto Calculation of Totals

Issues Found:

- Tax calculation formula missing.
- No shipping fee rules.
- Currency not specified.

FR11 – Checkout Requirements

Issues Found:

- Address format not defined.
- Supported regions/countries missing.
- Multi-address support not mentioned.

FR12 – Store Order & Confirmation

Issues Found:

- Content of confirmation message not defined.
- No confirmation email or receipt mentioned.
- No handling for failed order creation.

3.4 Order Management

FR13 – Order History

Issues Found:

- Missing details about what is displayed in order details.
 - No pagination or search filters mentioned.
 - No mention of filtering by date or order ID.
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FR14 – Update Order Status

Issues Found:

- Allowed status transition rules missing.
 - No audit logs defined.
 - No role-based permissions described.
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FR15 – Order Status Notifications

Issues Found:

- Notification channel not defined (email/SMS/in-app).
 - No message templates.
 - Notification history not mentioned.
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3.5 Payment Integration

FR16 – Payment Gateway Integration

Issues Found:

- Supported payment methods missing.
 - No mention of retry logic after payment failure.
 - No security requirements (PCI DSS, encryption).
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FR17 – Payment Failure Handling

Issues Found:

- Error message format not defined.
 - No centralized logging strategy.
 - No mention of handling pending/unknown payment states.
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3.6 Reporting & Analytics

FR18 – Sales Reports & Customer Summary

Issues Found:

- Types of reports not defined.
 - No visualization format specified (tables/charts).
 - No data refresh rate mentioned.
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FR19 – Export Reports

Issues Found:

- Export file structure undefined.
 - No size limits for export.
 - Unclear if all reports or only specific ones are exportable.
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2. Static Testing Defect Table

Requirement	Defect Identified
FR1	Missing email format & error messages
FR2	Password criteria not defined
FR3	Token expiration, refresh mechanism missing
FR4	Reset method not defined
FR5	Input validation rules missing
FR6	Product status values not defined
FR7	Missing pagination & sorting details
FR8	Search type & filter behavior
FR9	Quantity rules & stock scenarios missing
FR10	Tax formula & currency missing
FR11	Address format & regions not defined
FR12	Confirmation details missing
FR13	Missing details displayed in order history
FR14	No status transition rules
FR15	Notification channel & template missing
FR16	Supported payment methods not specified
FR17	Error message & logging unclear
FR18	Report types & formats missing
FR19	Export structure undefined

3. Requirements Quality Checklist

Clarity

- Several requirements are ambiguous or incomplete.
Needs improvement

Completeness

- Missing validation rules, formats, transitions, scenarios.
Incomplete

Consistency

- No conflicting requirements found, but lacking details decreases consistency.
Needs enhancement

Testability

- Many requirements cannot be tested without additional detail.
Not fully testable

Feasibility

- All requirements are feasible but need refinement.
Feasible with clarifications

Traceability

- No IDs missing; each requirement properly labeled FR1–FR19.
Good traceability
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4. Full Static Testing Report

Objective

The goal of the static testing session was to evaluate the SRS requirements for completeness, clarity, consistency, correctness, and testability before development or dynamic testing begins.

Scope

The review includes all 19 functional requirements across:

- Customer registration & authentication
- Product management
- Cart & checkout
- Order management
- Payment
- Reporting & analytics

Method Used

- Requirement reading & walkthrough
- Checklist-based review
- Identification of ambiguities and missing info
- Documentation of defects

Key Findings

- 80% of requirements lack essential details.
- Critical areas needing refinement: validation rules, error handling, statuses, formats, and workflow definitions.
- Tests cannot be designed for many requirements due to missing conditions.

Impact

Without refinement:

- Developers may implement incorrect behavior.
- QA cannot create accurate test cases.
- The system may fail to meet business expectations.

Recommendations

- Add validation rules for all input fields.
- Define standard error messages.
- Specify workflows for authentication, checkout, and orders.
- Document status transitions.
- Specify formats for exports, notifications, reports.
- Add role-based access control definitions.
- Provide examples and acceptance criteria for each requirement.